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B737 FLEET MAGAZINE



An exclusive fleet magazine for EAL 737 pilots.

Fleet News • Deeper look into B737 incidents and accidents • Breakthroughs in flight • Notices •
System reviews

NOTE

This magazine is for information purposes only. **USE COMPANY AND APPROVED MANUALS AS PRIMARY REFERENCE.**

Thank you for taking time to read The EAL 737 Fleet MAGAZINE. All readers are encouraged to submit their questions and suggestions, regarding this magazine to israelya@ethiopianairlines.com

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March 2026



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MESSAGE FROM CHIEF PILOT

Dear fellow pilots, warm greetings to all of you.

It has now been three months since our last newsletter update. Your motivation, professionalism, and commitment to safe operations continue to stand out—especially through what has been, as usual, a demanding season. Over the past months, we have also seen several important developments that I would like to share with you.

We have proudly welcomed **a new group of senior captains during the third-round recruitment**, bringing 12 experienced leaders into our team. Their willingness to take on additional responsibilities and their commitment to safety and operational excellence were evident throughout the process.

Following their arrival, we established a new focal team **Standardization**—dedicated to streamlining line operations and reinforcing a professional working environment. Their work is already underway, and you will begin to see improvements in the coming weeks.

Our fleet expansion is also progressing well. **We've received a new B737 MAX last month and assisted in the delivery of another B737 NG to Air Congo**, thanks to the dedication of Captain Dawit F. and Captain Israel Y. We expect additional aircraft deliveries before the end of the year, contributing significantly to our operational capacity.

We also underwent an IOSA audit, which remains a crucial element in ensuring the highest levels of operational safety. Despite our extensive preparation, the audit, as designed, identified few areas requiring improvement. We are now working diligently to address these gaps, and your support is essential. Adhering strictly to sterile cockpit procedures, recording all maintenance defects—no matter how repetitive—and consistently applying our policies and procedures enhance both our audit performance and our day-to-day safety.

An important operational milestone was achieved in our B737 MAX operation: **the transition from mandatory APU start during the ETOPS segment to an APU on demand procedure**. This change followed months of research into aircraft capabilities and close collaboration with the Maintenance Department. The result is significant cost savings through reduced fuel burn and extended APU life. Special appreciation is extended to Captain Dagmawi A. for his leading role in making this achievement possible.

Continued on next page ...

Innovation within our pilot group remains impressive. Many safety enhancing and cost saving ideas continue to surface. Captain Bereket proposed installing fins on the aft fuselage of the B737 to reduce vortex induced drag—an idea with the potential to save the airline millions annually. Another impactful idea by Captain Dagmawi aims at reducing ATC workload during peak hours. We hope to see these contributions implemented soon.

As we prepare for the rainy season in Addis, please review the GPS jamming/spoofing procedures in the FCOM and remain alert for reduced visibility and wet runway conditions. **To all instructors, I urge you to uphold the highest safety standards during training.** Do not permit unstabilized approaches, hard landings, or any other undesired aircraft condition. You serve as the final safety gate, and your vigilance is vital.

Finally, congratulations to all newly checked-out captains and first officers, and to those who have recently married or welcomed new babies. Our heartfelt condolences go to colleagues who have lost loved ones. Wishing you all safe flights and continued excellence.

**Safe Flight,
Capt. NATHAN ELIAS
Chief Pilot – B737&B767**



EAL B737

FLEET NEWS

As the B737 fleet continues to evolve and expand its operational capabilities, several new initiatives have recently been introduced to further strengthen efficiency, coordination, and operational consistency across the fleet. These developments reflect the airline's continued focus on improving operational processes while maintaining the high safety and performance standards expected from the operation. The following updates highlight some of the latest initiatives currently underway



1. Standardization Team

A new Standardization Focal Team has recently been formed with the primary objective of improving consistency and alignment across the entire B737 operation.

As the fleet grows and operations become increasingly complex, maintaining a high level of standardization across aircraft operations, training practices, and operational procedures becomes ever more important. The newly established team will focus on identifying areas where standardization can be strengthened and will work closely with other departments and focal groups to implement improvements where necessary.

The scope of the Standardization Team covers several important areas, including aircraft standardization, operational standardization, and training standardization.

By working alongside departments such as Training, Safety, and Maintenance, the team aims to ensure that operational procedures and practices are applied consistently throughout the fleet.

Ultimately, the goal of the Standardization Team is to support a more harmonized operational environment, reduce procedural variability, and contribute to a safer and more efficient B737 operation.



2.Outstation Focal Team

Another important development is the establishment of a dedicated Outstation Focal Team tasked with addressing operational issues that arise at stations away from the airline's home base.

Operations at outstations often involve coordination with a variety of local stakeholders, including ground handling providers, airport authorities, transportation services, and accommodation partners. When operational issues arise at these stations, resolving them can require careful follow-up and communication with multiple parties.

To streamline this process, the Outstation Team has assigned specific team members responsibility for particular stations within the network. When an issue occurs at a given outstation, flight crews or operational teams can reach out to the designated focal member responsible for that station. The focal member will then coordinate with the relevant stakeholders to help address and resolve the matter.

Issues handled by the team may include operational coordination challenges, crew transportation concerns, layover-related matters, ground handling issues, and other station-specific operational irregularities.

By creating a dedicated team focused on outstation operations, the airline aims to improve response times, strengthen communication channels and ensure that concerns raised by flight crews receive timely follow-up and resolution.

3.APU On-Demand Procedure for B737 MAX

A procedural update has also been implemented for the B737 MAX fleet regarding the operation of the Auxiliary Power Unit (APU) during Extended Diversion Time Operations(ETOPS).

Under previous procedures, the APU was required to be started during ETOPS segments as a precautionary measure to ensure its immediate availability if needed. However, following extensive operational analysis and demonstrated system reliability, the APU on the B737 MAX has achieved a 95 percent confidence level of reliability.



Based on this performance data, the procedure has now been revised so that starting the APU during ETOPS segments is no longer required as a standard practice. Instead, the APU will now be operated on an on-demand basis, meaning it will only be started when operational conditions require it.

This change provides several benefits to the operation. By eliminating unnecessary APU operation during cruise, the fleet can achieve measurable fuel savings while also reducing overall emissions. The procedure therefore contributes not only to improved operational efficiency but also to the airline's environmental sustainability efforts.

At the same time, the change reflects the high reliability and performance of the MAX APU system while maintaining the necessary safety margins for ETOPS operations.



MAX LRD SYSTEM

INTRODUCTION

The Load Reduction Device (LRD) is a mechanical safety feature installed in the CFM LEAP-1B engines on the Boeing 737 MAX.

It is designed to protect the engine structure and aircraft from excessive loads and vibration following severe engine damage such as a fan blade failure or major bird strike.

LRD is designed to reduce unbalanced dynamic loads transmitted to engine structures and pylon in case of a full or partial fan blade out event by decoupling the fan rotor.

The LRD activates automatically when the engine detects extreme vibration or severe fan damage, typically caused by:

- Fan blade separation
- Major bird strike
- Severe rotor imbalance
- Catastrophic fan damage



During LRD activation:

- Fan rotor partially disconnects from support structure
- Vibration loads to the engine and wing are reduced
- The fan may rotate slightly off-axis
- Engine internal seals or oil lines may open or dislodge

This protects the aircraft from structural loads but may lead to secondary effects

Investigations found a potential consequence after LRD activation:

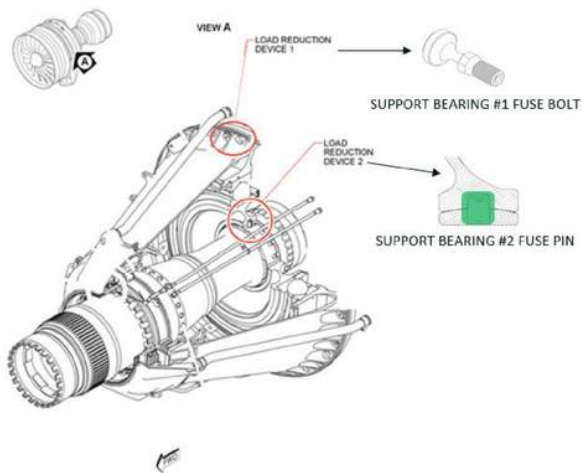
- Engine oil may leak into the compressor airflow
- Oil is exposed to hot engine sections
- Oil vapor becomes smoke or fumes
- Smoke may enter cockpit or cabin through the bleed-air system.

LRD activation itself does not generate a specific cockpit message.

However, the crew may observe:

- Engine damage indications
- High vibration
- Smoke or fumes in cockpit/cabin
- Possible engine failure indications

In some events, crews noticed smoke shortly after bird strike and engine damage



If LRD activation is suspected (e.g., after severe engine damage):

Priorities:

- Don oxygen masks and smoke goggles if smoke present
- Run appropriate non-normal checklist
- Consider engine shutdown
- Plan diversion/return

In reported events, pulling the engine fire switch stopped the smoke source by closing bleed air valves.

Why it is a safety concern?

- Cockpit visibility reduction during critical phases (takeoff/landing).
- High pilot workload during already stressful engine failure scenarios.
- Pilots may not immediately recognize the cause of smoke.
- Some pilots may not yet be aware of this failure mode.

Current Mitigations

Boeing Operational Changes, Boeing updated:

- FCOM system description
- QRH procedures, Now recommending that engine failure with smoke/fumes should trigger the ENGINE FIRE or Engine Severe Damage or Separation checklist.

Planned Technical Fix

CFM and Boeing are developing a software update that will:

- Automatically close the Pressure Regulating Shutoff Valve (PRSOV) when the LRD activates.
- This prevents bleed air (and smoke) from entering the aircraft



BIRD STRIKE

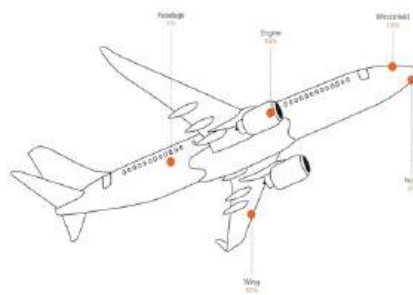
Prevention, Responses And Reporting

1.0 WHAT IS A BIRD STRIKE?

A bird strike is strictly defined as a collision between a bird and an aircraft which is in flight or on a take off or landing roll. The term is often expanded to cover other wildlife strikes - with bats or ground animals.

Bird strikes may occur during any phase of flight but are most likely during the take-off, initial climb, approach and landing phases due to the greater numbers of birds in flight at lower levels. Since most birds fly mainly during the day, most bird strikes occur in daylight hours as well.

2.0 LOCATIONS OF BIRD STRIKE



3.0 COMMON MISCONCEPTIONS

- Birds don't fly at night.
- Birds don't fly in poor visibility, such as in clouds, fog, rain, or snow.
- Birds can detect airplane landing lights and weather radar and avoid the airplane.
- Airplane colors and jet engine spinner markings help to repel birds.
- Birds seek to avoid airplanes because of aerodynamic and engine noise.
- Birds dive to avoid an approaching airplane.

4.0 Preventive Strategies

Airports are responsible for bird control and should provide adequate wildlife control measures.

If large birds or flocks of birds are reported or observed near the runway, the flight crew should consider:

- Delaying the takeoff or landing when fuel permits.
- Advise the tower and wait for airport action before continuing.
- Take off or land on another runway that is free of bird activity, if available.
- Discuss bird strikes during takeoff and approach briefings when operating at airports with known or suspected bird activity.
- Be extremely vigilant if birds are reported on final approach. If birds are expected on final approach, plan additional landing distance to account for the possibility of no thrust reverser use if a bird strike occurs.
- Below 10,000 feet, keep speed below 250 knots if operationally possible.
- Below 2,000 feet, climb at the maximum rate to reduce the flight time exposure to a strike hazard.
- Descend with idle power and avoid extended lowaltitude level flight, particularly over water courses, reserves, or other areas of known or expected bird activity.
- When landing is assured, consider landing through birds versus a missed approach to avoid birds. This reduces the energy of the collision, the potential for increased damage associated with engines at a high power level, and the potential for multiple engine ingestions at low airplane energy states and low altitude.
- Avoid or minimize maneuvering at low altitude to avoid birds.

BIRD STRIKES DURING TAKEOFF ROLL

If a bird strike occurs during takeoff, the decision to continue or reject the takeoff is made using the criteria found in the rejected takeoff maneuver of the QRH. If a bird strike occurs above 80 knots and prior to V₁, and there is no immediate evidence of engine failure (e.g., failure, fire, power loss, or surge/stall), the preferred option is to continue with the takeoff followed by an immediate return, if required.

5.0 DETECTING A BIRD STRIKE WHILE IN FLIGHT

- **Visual:** Birds seen in close proximity to the airplane or colliding with the airplane, bird remains on windshield, cracked windshield.
- **Tactile:** Vibration of airframe or engine, thrust loss, asymmetric thrust, increased drag, abnormal airplane handling characteristics.
- **Auditory:** Noise of strike or noise attributed to resulting damage: engine surging, compressor stalls, aerodynamic noise from damaged radome, loss of pressurization from pressure vessel penetration.
- **Olfactory:** Smoke, odor, or cooked bird smell.
- **Engine Indications:** reduction or fluctuation in primary power parameter (e.g., engine pressure ratio, fan speed, or equivalent), abnormal fuel flow, abnormal engine vibration, engine failure, engine exceedances.
- **Flight Instruments:** loss of data or erroneous indications arising from damage to air data sensors or angle of attack sensors.
- **Other airplane systems or structure affected directly by a strike:** Damaged communications or navigation antennas, damage to exposed electrical wiring or hydraulic lines, damaged radome or weather radar, broken landing lights, or cascading and multiple effects from sensor damage or engine damage.

6.0 RESPONSES TO A KNOWN OR SUSPECTED BIRD STRIKE

Immediate action:

- Fly the airplane and maintain flight path control.
- monitor flight and engine instruments.

Multiple engine failure or thrust loss

- attempt to restart engine(s).

Severe engine damage

- Shut down engine according to procedure.

Strong engine vibration

- reduce thrust, which will often reduce vibration.
- Follow QRH guidance.

Multiple engine ingestion and abnormal engine indications

- Consider air turn-back or diversion to nearest suitable airport.

Known or suspected airframe damage or engine damage

- maintain or reduce speed – do not accelerate unless necessary for safety of flight or to maintain flight path control.

Damaged windshield or depressurization

- below 10,000 feet, discontinue climb and level off.
- above 10,000 feet, descend to 10,000 feet or the minimum safe altitude.

Known or suspected strike with landing gear extended or in takeoff or landing configuration with high lift deployed

- use available system information to assess possible damage to flight controls and high lift devices, and make minimal and prudent changes in airplane configuration in accordance with the flight phase.
- use available system information to assess possible damage to landing gear and associated systems, including exposed electrical, pneumatic, and hydraulic systems, and potential effects on the ability to steer and stop on the runway.

Known or suspected strikes to air data and angle of attack sensors

- be aware that this may affect other airplane systems and have cascading effects.
- be aware of the potential for loss or erroneous air data and degraded flight control modes, including loss of envelope protection or limiting, unreliable airspeed, propulsion systems in alternate mode.

Bird strikes during approach or landing

- if the landing is assured, continuing the approach to landing is the preferred option. if more birds are encountered, fly through the bird flock and land.
- maintain as low a thrust setting as possible.
- if engine ingestion is suspected, limit reverse thrust on landing to the amount needed to stop on the runway. reverse thrust may increase engine damage, especially when engine vibration or high exhaust gas temperature is indicated.

7.0 Bird Strike Reporting

Whilst Bird Strike Reporting is a reactive response to the potential hazard, there is no alternative widely-available means of monitoring potentially hazardous bird activity and this has been recognized by Regulatory action to improve the extent of reporting.

What to Report

The information that is commonly sought in a bird strike report includes:

- General information (flight number or registration, aircraft and engine type)
- Location and time, (geographic location or airport/runway used, time, IAS and height (AGL) during the occurrence, light conditions - day , night, twilight)

- Flight parameters - Altitude or height AGL; IAS
- Phase of flight (take-off roll, initial climb, climb, cruise, descent, approach, landing roll)
- Part(s) of aircraft struck and (if) damaged (engine and its number, radome, windshield, landing gear, fuselage, tail, etc)
- Effects on flight (rejected take off, emergency/precautionary landing, engine(s) shut down and other consequent effects)
- Relevant meteorological conditions (visibility, cloud cover, precipitation)
- Species information (exact species or species group if known)
- Number of birds seen, and number struck
- Size of birds - small, medium and large if not identified to species or species group)
- Prior warning of bird activity by RTF/ATIS/NOTAM/BIRDTAM
- Any other relevant information and remarks regarding the occurrence

As per the ECAA (Ethiopian Civil Aviation Authority) Wildlife and Bird Strike Reporting Procedure, the pilot in command will be responsible to report any strikes to ATC as soon as bird strike encountered. ATC may learn of a strike by radio reports from pilots and disseminate the information to Airport wildlife management or safety offices.

Accordingly all captains are required to report bird strike or bird activity to ATC immediately when workload permits and file bird strike report form to flight operations through Logipad.



PERFORMANCE BASED NAVIGATION

THE FOUNDATION OF MODERN AIR NAVIGATION

1.Introduction

Modern airspace is becoming increasingly congested, and traditional navigation methods based primarily on ground-based navigation aids are no longer sufficient to meet the demands of today's global aviation system. To address this challenge, the aviation industry has progressively transitioned toward Performance-Based Navigation (PBN), a concept developed by the International Civil Aviation Organization (ICAO) to improve the efficiency, predictability, and safety of air navigation.

PBN represents a shift from sensor-based navigation to performance-based navigation specifications, allowing aircraft to fly more precise and predictable flight paths. Today, PBN forms the backbone of modern instrument procedures, including many Standard Instrument Departures (SIDs), Standard Terminal Arrival Routes (STARs), and approach procedures used worldwide.

2.From Sensor-Based Navigation to Performance Based Navigation

Historically, aircraft navigation relied heavily on specific ground-based navigation aids such as VOR, DME, and NDB. Under this system, routes and procedures were designed around the physical location of these navigation aids, often resulting in indirect routing and inefficiencies.

Performance-Based Navigation introduced a new philosophy. Instead of requiring a particular navigation sensor, PBN defines the required navigation performance that an aircraft must achieve within a specific airspace.

In other words, rather than asking "Which navigation equipment is being used?", PBN focuses on "How accurately can the aircraft navigate?"

This concept allows aircraft equipped with modern avionics and satellite navigation systems to fly more direct and optimized routes, independent of the location of ground navigation stations.

3.Core Components of PBN

The PBN framework is built around three key elements:

A.Navigation Specifications

Navigation specifications define the performance requirements that aircraft must meet to operate in a given airspace or procedure. These specifications include **accuracy, integrity, continuity, and functionality** requirements.

Two major categories exist:

- 1.RNAV (Area Navigation)
- 2.RNP (Required Navigation Performance)

Both RNAV and RNP allow aircraft to navigate between waypoints rather than relying on direct tracking of ground-based navigation aids.

B.Navigation Infrastructure

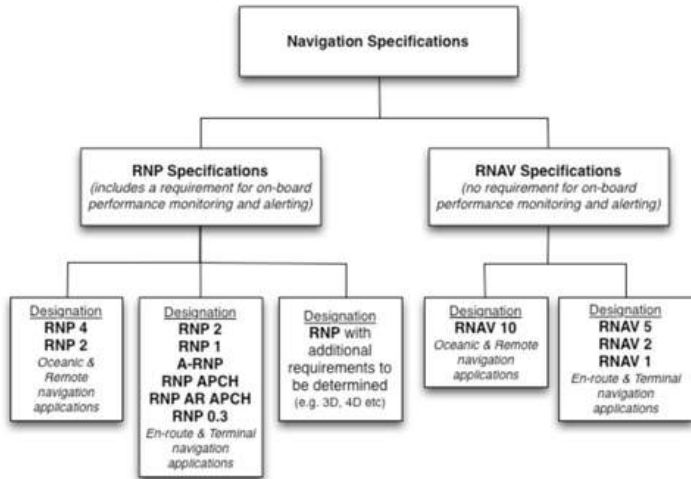
PBN relies on a combination of navigation sensors, including:

- Global Navigation Satellite Systems (GNSS)
- Inertial Reference Systems (IRS)
- Distance Measuring Equipment (DME/DME)
- Other onboard navigation systems

These sensors feed data to the aircraft's Flight Management System (FMS), enabling precise navigation along defined flight paths.

C.Navigation Applications

Navigation applications refer to the specific procedures and airspace concepts that use PBN capabilities. These include advanced departure, arrival, and approach procedures that increase efficiency and airspace capacity.



3.RNAV and RNP: Understanding the difference

Although the terms RNAV and RNP are often used interchangeably, they represent two distinct types of navigation specifications.

RNAV (Area Navigation) allows an aircraft to fly on any desired flight path within the coverage of navigation aids or satellite signals. RNAV specifications define the required lateral navigation accuracy but do not require onboard monitoring of navigation performance.

For example:

- RNAV 5
- RNAV 2
- RNAV 1

These numbers refer to the required navigation accuracy in nautical miles. RNAV 1, for instance, means the aircraft must maintain a navigation accuracy of within 1 nautical mile for at least 95 percent of the flight time.

RNP (Required Navigation Performance) builds upon RNAV by adding onboard performance monitoring and alerting capabilities. This means the aircraft's navigation system continuously monitors its own performance and alerts the crew if the required navigation accuracy cannot be maintained.

Examples include:

- RNP 4
- RNP 2
- RNP 1
- RNP APCH
- RNP AR APCH

Because of its additional monitoring capability, RNP enables more precise procedures, including curved flight paths and approaches in challenging terrain environments.

4.Operational Benefits of PBN

The widespread implementation of PBN has brought numerous advantages to modern aviation operations.

-Improved Airspace Efficiency

PBN enables the design of more direct and optimized flight paths, reducing track miles and minimizing unnecessary vectoring by air traffic control. This improves traffic flow and allows airspace to accommodate a greater number of aircraft safely.

-Fuel Savings and Environmental Benefits

By enabling shorter routes and more efficient climb and descent profiles, PBN helps reduce fuel consumption and lower aircraft emissions. Continuous descent operations, which are often supported by PBN procedures, further reduce fuel burn and noise levels around airports.

-Enhanced Safety

Precise navigation performance allows aircraft to maintain predictable flight paths, reducing the risk of navigation errors. RNP procedures in particular provide enhanced safety through onboard monitoring and alerting.

-Access to Challenging Airports

In areas where terrain or airspace constraints limit the use of traditional procedures, PBN allows the design of advanced instrument procedures that provide safe and reliable access to airports that may otherwise be difficult to serve.

5.PBN in Daily Airline Operations

For airline pilots, PBN is already a routine part of everyday operations. Many modern SIDs, STARs, and approach procedures are designed using RNAV or RNP specifications, and aircraft such as the Boeing 737 are fully equipped to operate in PBN environments.

The Flight Management System plays a central role in executing PBN procedures, guiding the aircraft along predefined lateral and vertical flight paths. However, pilots must remain vigilant in ensuring proper procedure selection, verifying navigation database accuracy, and monitoring aircraft performance throughout the flight.

Understanding the underlying concepts of PBN also helps pilots appreciate the reasoning behind many modern airspace procedures and supports better situational awareness during complex terminal operations.

6.The Future of Air Navigation

Performance-Based Navigation continues to be a cornerstone of ICAO's global air navigation strategy. As satellite navigation technology and aircraft avionics continue to advance, PBN will play an even greater role in enabling efficient and sustainable air traffic management worldwide.

For flight crews, this evolution represents a transition toward more precise, predictable, and efficient navigation, supporting safer operations and a more capable global aviation system.

A-VDGS

SAFEDOCK Advanced Visual Docking Guidance System (A-VDGS)

Overview

The Safedock Advanced Visual Docking Guidance System (A-VDGS) is designed to provide active guidance to both pilots - using alphanumeric, azimuth and closing rate indicators - to support safe, efficient and precise aircraft parking at the gate during all operating conditions. Safedock uses an eye-safe infrared laser and 3D scanning technique to detect the aircraft and measure nose, engine and wing positions to verify aircraft type and safely guide the aircraft to the correct stop position.

Pilot Responsibilities:

When parking with guidance from the Safedock A-VDGS, pilots are asked to adhere to the following safety procedures:

- never enter the gate area unless the correct aircraft type and running arrows are displayed on the pilot display;
- never enter the gate area unless you can see that the safety zone is clear and it is safe to enter;
- never move the nose of the aircraft past the jet bridge unless the A-VDGS is displaying the guidance screen with centerline indicator;
- always stop the aircraft when the A-VDGS displays a WAIT or STOP message. The Automated Docking Process:

A-VDGS capture, tracking & guidance:

When the A-VDGS captures the incoming aircraft the guidance screen appears:

- vertical yellow bar represents the centerline and provides closing rate to stop position;

- floating yellow arrow indicates position of the aircraft relative to centerline;
- red flashing arrow indicates the direction the pilot should steer to center on centerline;
- the absence of a red direction arrow indicates the aircraft is on the centerline;
- a digital countdown is provided for the last 30 meters.

Safety procedures:

a. General warning.

The SAFEDOCK DGS has a built-in error detection program to inform the aircraft pilot of impending dangers during the docking procedure. If the pilot is unsure of the information, being shown on the DGS display unit, he/she must immediately stop the aircraft and obtain further information for clearance.

b. Item to check before entering the stand area.

WARNING: The pilot shall not enter the stand area, unless the docking system first is showing the vertical running arrows. The pilot must not proceed beyond the bridge, unless these arrows have been superseded by the closing rate bar.

The pilot shall not enter the stand area, unless the aircraft type displayed is equal to the approaching aircraft. The correctness of other information, such as DOOR 2, shall also be checked.

c. The SBU message.

The message STOP SBU means that docking has been interrupted and has to be resumed only by manual guidance. Do not try to resume docking without manual guidance.

SLOW: If the aircraft is approaching faster than the accepted speed (> 4 knots), the system will show a SLOW message requiring pilots to adjust their speed. When the A-VDGS is in low visibility mode during heavy fog, rain or snow, the SLOW message is displayed requiring pilots to adjust their speed.

If the aircraft approaches with a speed higher than the A-VDGS can handle, a **STOP** message is displayed. The pilot must stop until the A-VDGS is restarted or manual guidance is provided.

When the aircraft reaches the stop position, the AVDGS indicates STOP. (There may be variations in the STOP position until pilots are acclimated.)

An **OK** message indicates the docking is complete.

If the aircraft over shoots the stop position by more than one meter a **TOO FAR** message is displayed. A TOO FAR message may require the aircraft to be moved back.

Ground staff chock aircraft wheels and the A-VDGS displays **CHOCK ON** message.

NOTE: After the main gear have been chocked, the technician must notify the PIC "chocks inserted".

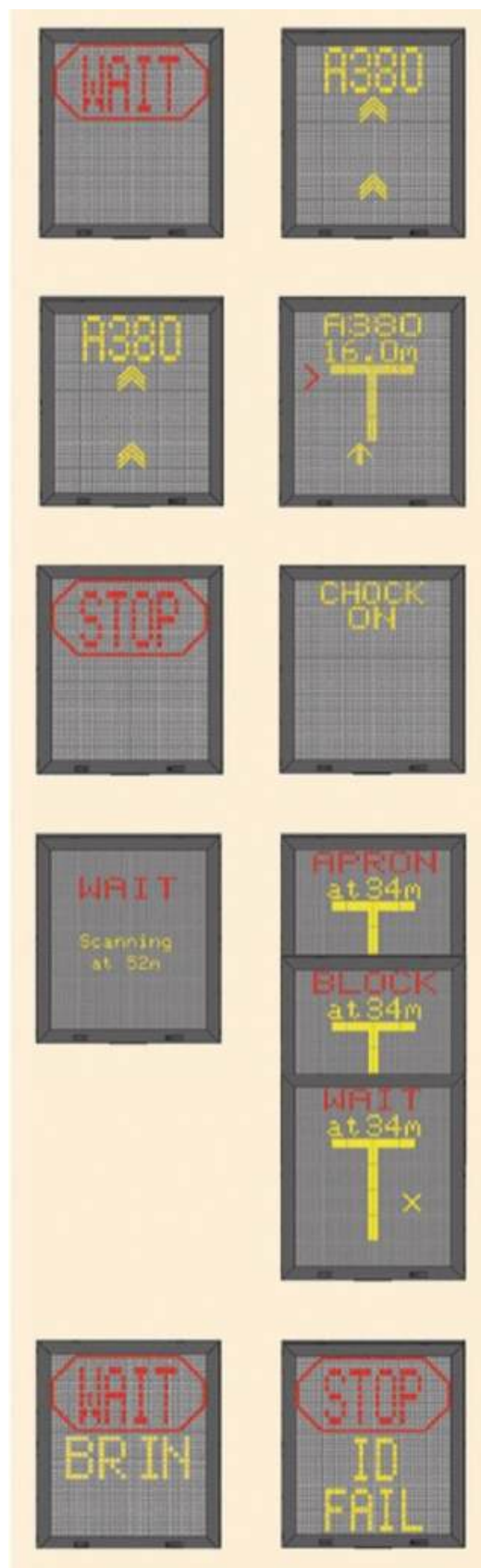
APRON BLOCK: An APRON BLOCK occurs if the A-VDGS detects an obstacle blocking its view of the aircraft. The pilot is halted with a **WAIT APRON BLOCK** message. When the obstacle is removed the docking will restart on its own. The pilot should not proceed past the bridge until the guidance screen is displayed.

BR IN: A BR IN message occurs when the boarding bridge is inside the safety zone.

The pilot is halted with a **WAIT BR IN** message. When the bridge is moved to its parked position the docking will restart on its own. The pilot should not proceed until the guidance screen is displayed.

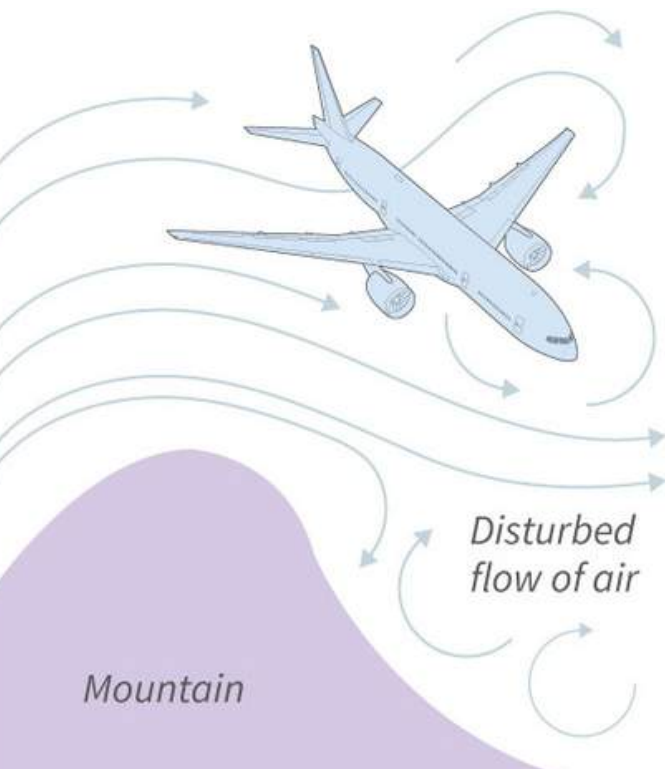
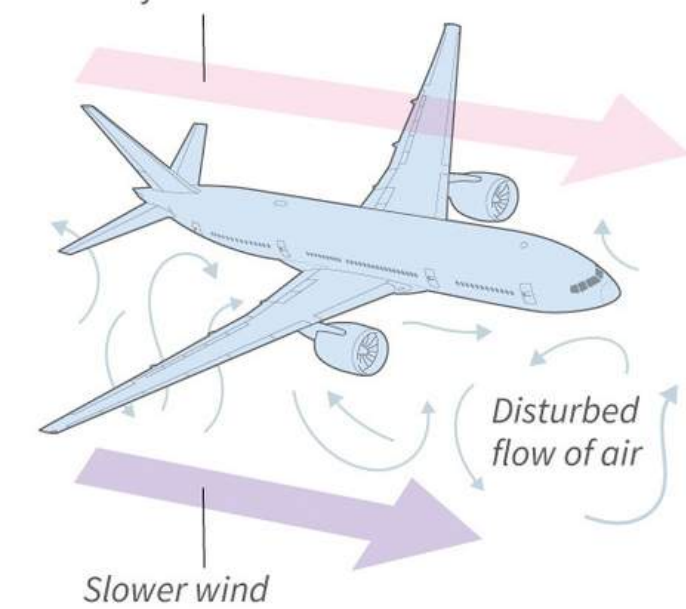
ID FAIL: An ID FAIL occurs when the A-VDGS cannot confirm verification of the aircraft type.

If verification is not made 12 meters before the stop position, the A-VDGS will display WAIT and make a second attempt at verification. If this fails, the STOP ID FAIL message will display. Pilots will require manual guidance to the gate.



CLEAR AIR TURBULENCE

Staying Ahead Of The Invisible



Clear Air Turbulence (CAT) refers to sudden, unpredictable turbulence occurring in clear skies, typically at cruising altitudes. It is most commonly associated with strong wind shear near jet streams.

Unlike turbulence associated with convective weather, CAT:

- Occurs without visible clouds
- Is not detectable by onboard weather radar
- Can develop rapidly and dissipate quickly

For flight crews, CAT often presents as sudden jolts or sustained moderate turbulence during otherwise smooth cruise conditions.

CAT typically occurs in regions with significant wind speed gradients including:

1. Jet Stream Boundaries

One of the most common places to encounter Clear Air Turbulence (CAT) is near jet streams.

CAT is usually found around the edges of the jet stream, particularly:

- Above the jet core
- Below the jet core
- On the cold side of the jet stream

2. Mountain Wave Regions

Even when skies appear clear, strong winds over mountain ranges can create standing waves and rotor zones that extend into cruise altitudes

3. Tropopause Variations

Sharp changes near the tropopause boundary can create shear layers where CAT forms.

4. Upper-Level Frontal Systems

Strong temperature gradients at altitude can generate turbulence even when the atmosphere appears calm.

Why is CAT a safety concern?

CAT can affect operations in several ways:

1. Passenger and Cabin Safety

Unexpected turbulence is one of the leading causes of in-flight injuries, particularly when passengers or cabin crew are not seated

2. Structural Loads

Modern airliners are designed with large structural margins, but repeated turbulence exposure can increase structural stress and maintenance considerations.

3. Flight Path Control

In moderate to severe CAT:

- The autopilot may disconnect due to excessive control inputs required.
- The aircraft may experience sudden altitude or attitude deviations.
- Pilots may need to hand fly temporarily until conditions stabilize.

When encountering moderate or greater turbulence, maintaining the appropriate turbulence penetration speed helps reduce structural loads.

Typical guidance includes:

- Avoid large control inputs
- Maintain a steady attitude
- Allow altitude variations rather than aggressively chasing altitude
- Flight crew should refer to QRH and aircraft-specific guidance for recommended speeds depending on aircraft weight and configuration

Detection and Forecasting

Since CAT is invisible to radar, pilots rely on forecasting tools and pilot reports (PIREPs).

Useful Tools include:

- SIGMETs and AIRMETs
- Wind and Temperature Charts
- Turbulence Forecast Models
- PIREPs

Notable Clear Air Turbulence Incidents in Airline Operations

1. Singapore Airlines Severe CAT Event (2024)

One of the most widely reported CAT events in recent years occurred in May 2024 involving a Boeing 777-300ER operating from London to Singapore.

The aircraft, cruising at FL370, encountered sudden severe clear air turbulence, resulting in a rapid altitude fluctuation within seconds. During the event, several passengers and cabin crew sustained injuries, prompting the flight crew to divert to Bangkok to obtain medical assistance for those affected.

2. United Airlines CAT Encounter over the United States (2019)

A United Airlines Boeing 737 encountered unexpected clear air turbulence while cruising over the central United States.

During cruise in clear skies, the aircraft encountered sudden moderate to severe turbulence lasting about 20–30 seconds, likely caused by strong upper-level wind shear near the jet stream. As a result, several passengers who were not wearing seat belts sustained injuries, although the aircraft continued and landed normally at its destination

These events highlight an important operational insight: many turbulence-related injuries occur when passengers are not seated or their seat belts are not fastened, even when the turbulence lasts only a short time.

Experienced flight crew often use several practical strategies when CAT is suspected.

- Review upper wind charts during preflight and identify jet stream crossings along the route.
- Request ride reports from aircraft ahead and share turbulence encounters with ATC.
- Adjust Altitude Strategically: Often a 2,000–4,000 ft altitude change can significantly improve ride quality.
- If turbulence is expected, turn the seat belt sign on early and inform the cabin crew.
- Avoid over-controlling the aircraft. In turbulence, attitude stability matters more than precise altitude control



CONFLICT MANAGEMENT

THE ROLE OF CRM

1.0 Introduction

All ETHIOPIAN flight operations are based on the optimum use of Crew Resource Management. The principle of continuous mutual briefing and assistance shall be applied at all times.

In normal cockpit work the Captain shall establish open communication between crew members in the cockpit and in the cabin as well as with ground personnel and Air Traffic Services.

CRM teaches crew how to handle disagreements professionally, using Interpersonal skills, while keeping safety first.

2.0 Interpersonal competencies

Communication: Crew members must share information clearly so everyone understands the situation and can make good decisions and avoid conflicts even before they arise.

Conflict Management - To enhance bridge the communication gap, the aviation industry emphasizes conflict management strategies that involve addressing conflicts, achieving a rational climate, and avoiding intimidation, while adopting **assertive behavior, accepting criticism, and discouraging competition among** crew members to foster collaboration.

Leadership and Teamwork: The Captain is the designated leader. Lead by the Captain, the crew can expect to achieve a safe and efficient outcome in a climate that is rational and free of intimidation. Social interaction conflicts have to be addressed and managed. Each crew member takes the initiative to be an active and constructive part of the team.

Command Ability: the captain takes the lead to establish goals, control outcomes, and apply necessary corrections, while also evaluating and considering conditions suggested by other crew members.

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Command Ability: the captain takes the lead to establish goals, control outcomes, and apply necessary corrections, while also evaluating and considering conditions suggested by other crew members.

Team Ability: To further enhance collaboration and bridge the communication gap, fostering team ability is paramount; this requires crew members to act as constructive team members who take initiative, encourage cooperation, and support their colleagues, while actively seeking ideas, presenting their own views, providing appropriate feedback, and proposing alternative ideas when necessary.

3.0 Interdisciplinary Conflict Dynamics

3.1 Flight Crew vs. Maintenance Technicians

Conflicts often arise during the pre-flight phase regarding the interpretation of the Minimum Equipment List (MEL). A technician may believe a system is serviceable following a reset, while the Captain may harbor concerns about intermittent faults.

CRM Resolution: Utilize **Information Sharing for Decision Making**. Both parties must present clear, transparent data. The goal is not to "win" the argument but to reach a safe, informed decision based on the technical status of the aircraft.

3.2 Flight Crew vs. Cabin Crew

During emergency or high-workload phases, the flow of information can become a point of friction. The cabin crew may feel ignored during a medical emergency, while the flight crew is focused on a complex approach.

CRM Resolution: Establishing Open Channels of Communication. The Captain is responsible for setting a tone that encourages cabin crew to provide proactive, constructive support. This prevents "communication gaps" that can lead to missed safety cues.

4.0 Strategic Conflict Management Framework

To maintain flight safety, crews must adopt a structured approach to disagreement:

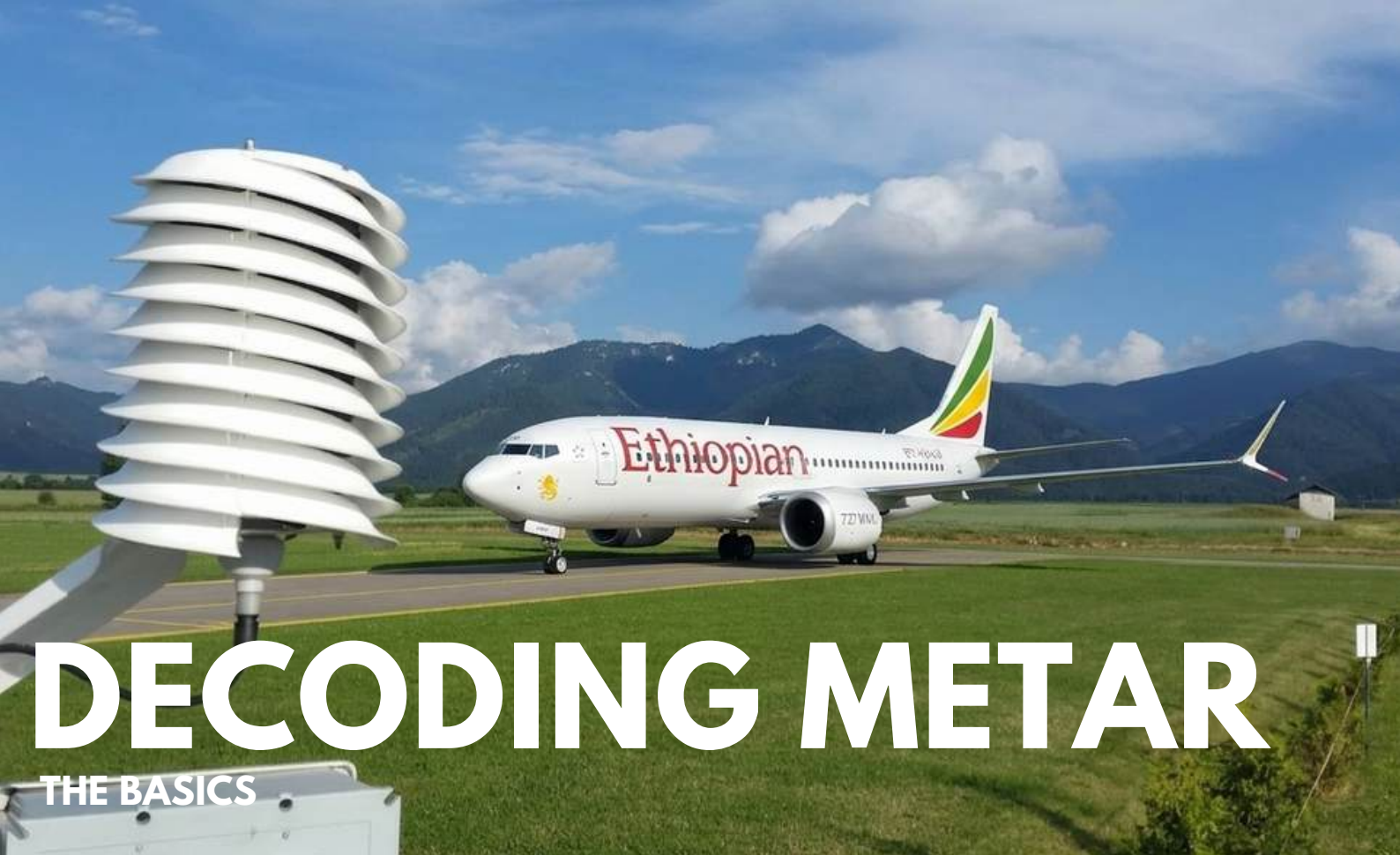
4.1 Create a Rational Climate: Strategies must be employed to avoid intimidation and discourage internal competition. A professional environment allows for "Professionalism Over Discord," where safety protocols are never compromised by personal differences.

4.2 Continuous Mutual Briefing: Effective CRM requires the constant exchange of assistance and information between all units—cockpit, cabin, and ground personnel.

4.3 Assertive Behavior and Feedback: Crew members at all levels are encouraged to adopt assertive behavior—stating concerns clearly and firmly—while accepting criticism and providing constructive feedback.

Conclusion

Conflict is an inherent risk in the high-pressure environment of aviation. However, by adhering to the **Ethiopian Airlines Standards of CRM**, teams can bridge the communication gap, avoid or resolve conflicts professionally when they arise.



DECODING METAR

THE BASICS

1.0 Introduction

The letters METAR stand for METeorological Aerodrome Report. METARs contain coded messages pertaining to the actual weather conditions at a given aerodrome, at a stated time.

METARs are usually issued **every half hour during aerodrome operating hours**. The aim of this article is to explain the METAR coding, group by group, to especially help understand the codes and clear some misconceptions.

METAR HAAB 231020Z 26012G25KT 220V300

(a) (b) (c) (d) (e)

Report Type- The first code, (a), is the identification of the type of report; in this case a METAR.

Aerodrome- The four-letter ICAO designator of the issuing aerodrome, (b);

Date-Time Group- The third group, (c), is the date/time group, which simply gives the date of the actual weather observation. The first two digits represent the day of the month, followed by the time in hours and minutes. Time is always given as Coordinated Universal Time (UTC),

Wind Information- The next item in the METAR (d) is the observed wind information. Firstly, the direction of the wind given in degrees true, rounded up or down to the nearest 10 degrees, and then the wind speed in knots, **which is a mean speed taken over a 10 minute period**. However, if a gust is observed which is at least 10 knots more than the mean wind speed, then a gust figure, comes after the mean wind.

Variability of direction is included when the wind direction, over the preceding 10 minutes, has changed by 60° or more. The letter **V** will appear between these two extremes., as depicted in (e).

If there is no wind, the coding, 00000KT, will be used. If the wind direction cannot be defined then **VRB (for variable)** replaces the direction.

Visibility-Visibility in the METAR is represented by the next group, depicted in red in the next example. **In the METAR, the reported visibility is the prevailing visibility and, may, under certain conditions, include the minimum visibility.** Here, the prevailing visibility is reported as 0800 metres.

Prevailing visibility is the visibility value which is either reached, or exceeded, around at least half the horizon circle, or within at least half of the surface of the aerodrome.

If the visibility in one direction, which is not the prevailing visibility, is less than 1500 m, or less than 50% of the prevailing visibility, the lowest visibility observed, and its general direction, should also be reported.

Up to 10 km, the visibility is measured in metres. For example, 6000 means that the prevailing visibility is 6000 metres. Once the visibility reaches 10 km or more, the code figure used is 9999.

Visibility of less than 50 metres is indicated by the code 0000. In this example the prevailing visibility is 800 metres.

METAR HAAB 231020Z 26012KT 0800

In some instances, runway visibility information is given in a METAR; this is known as Runway Visual Range (RVR.) **RVR is given only when either the horizontal visibility or the RVR, itself, is less than 1500 metres.** The RVR group starts with the letter R, and then goes on to give the runway in use, followed by the threshold visibility in metres.

In the following example, for Bole International, we have a prevailing visibility of 800 metres, with an RVR, at the threshold of Runway 25L, of 1100 metres.

**METAR HAAB 211020Z 26012G25KT 0800
R25L/1100**

If the RVR is more than the maximum reportable value of 1500 metres, the code P is used in front of the visibility value, **R25L/P1500**. If the visibility is less than 50 m then the prefix M will be used e.g. **R25L/M0050**

A letter can sometimes come after the RVR to indicate any trends that the RVR has shown. A **U** means that the visibility has increased by 100 m or more in the last 10 minutes, e.g. **R25L/1100U**. A **D** shows that visibility has decreased by 100 m or more in that same time period, **R25L/1100D**. An **N** added to the visibility group shows that there is no distinct trend observed, **R25L/1100N**.

The Weather Group- The next section of the METAR is the weather group. The weather group gives information on the present weather at, or near, the aerodrome at the time of the observation. The weather group +SHRA added to a METAR for example means "heavy showers of rain".

Thunderstorms- A Thunderstorm report will appear in a METAR if thunder has been heard within the last 10 minutes.

A thunderstorm is represented by the letters TS. If there is no precipitation, the letters TS will appear on their own. However, if there is precipitation, a further two letters, which signify the type of precipitation, are inserted after the TS. For example, if there is rain observed from the thunderstorm, TSRA will appear in the METAR. If hail were to be observed, the code would read TSGR, or TSGS, with GS meaning small hail.

Cloud Coverage- The next code-group to appear in the METAR gives detail of cloud coverage, as highlighted in red below. In this case the highlighted code means: overcast sky, base 2000 ft, with cumulonimbus.

METAR HAAB 211020Z 26012G25KT 0800 R30/1100 +SHRA OVC020CB

There are several prefixes which are used to describe cloud amount, at any given level. Cloud coverage is reported in the METAR using the following three-letter codes:

- **FEW (FEW)** meaning one to two eighths of cloud coverage.
- **SCATTERED (SCT)** meaning three to four eighths of cloud coverage.
- **BROKEN (BKN)** meaning five to seven eighths of cloud coverage.
- **OVERCAST (OVC)** meaning complete cloud coverage, or eight eighths.

TREND, BECMG, TEMPO- A TREND forecast is valid for 2 hours after the time of the observation of the METAR, and constitutes the final section of the METAR.

The change in weather conditions indicated by the code, TREND, can be further qualified by the codes, BECMG, meaning becoming, or TEMPO meaning temporarily.

BECMG indicates that the change in the present weather will be long-lasting. TEMPO, on the other hand, means that the change is temporary, and that the different conditions will prevail for periods of less than one hour, only, and no more than half the time period, in aggregate. The codes may be followed by a time period in hours and minutes. The time periods given may be preceded by FM meaning from, TL meaning until, or AT meaning at.

For example, TEMPO FM1020 TL1220 1000 +SHRA translates as: temporarily, from 1020Z to 1220Z, the visibility will reduce to 1000 metres, in heavy showers of rain.

If there is no expected change in the meteorological conditions being forecast by the METAR, the code NOSIG is used to indicate that **no significant change is expected in the next two hours**.

METAR LTFM 231020Z 26012G25KT 220V300 9999 -RA FEW060 SCT120 10/M02 Q0991 RETS WS ALL RWY NOSIG

FEW (1-2/8)



SCT (3-4/8)



BKN (5-7/8)



OVC (8/8)



Thank You!